

DS Practical - 4

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a. Write a program to: a. Implement a queue using an array.

CODE :

```
from array import *
import time

que = array('i', [])

def enqueue(data):
    que.append(data)
    print("Data inserted:", data)

def dequeue():
    if len(que) == 0:
        print("Queue is empty")
    else:
        print("Data removed:", que.pop(0))

def peek():
    if len(que) == 0:
        print("Queue is empty")
    else:
        print("First element is:", que[0])

def show():
    print("Queue:", que)

# Separate list for scheduling
jobs = [("P1", 2), ("P2", 1), ("P3", 2), ("P4", 4)]

print("-----Running Order-----")
print(jobs)

while jobs:
    job, timer = jobs.pop(0)
    print(f"Running: {job} ({timer} sec)")
    time.sleep(timer)

enqueue(10)
enqueue(20)
enqueue(30)

show()
peek()
dequeue()
show()

print("KOMAL S049")
```

OUTPUT :

```
-----Running Order-----
[('P1', 2), ('P2', 1), ('P3', 2), ('P4', 4)]
Running: P1 (2 sec)
Running: P2 (1 sec)
Running: P3 (2 sec)
Running: P4 (4 sec)
Data inserted: 10
Data inserted: 20
Data inserted: 30
Queue: array('i', [10, 20, 30])
First element is: 10
Data removed: 10
Queue: array('i', [20, 30])
KOMAL S049
```

b. Simulate a simple queuing system (e.g., customer service queue).

CODE :

```
import time
queue = []

queue.append("Komal")
queue.append("Ayush")
queue.append("Parth")

print("Processing Tickets")
print(queue)

while queue:
    customer = queue.pop(0)
    print(f"Processing: {customer}")
    time.sleep(2)

print("KOMAL S049")
```

OUTPUT :

```
Processing Tickets
['Komal', 'Ayush', 'Parth']
Processing: Komal
Processing: Ayush
Processing: Parth
KOMAL S049
```